

Boss Challenge — Paper 24 (Class 7)

Nutrition in Animals | Physical & Chemical Changes | Fractions & Decimals | Comparing Quantities
One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The mode of nutrition in which an animal takes in solid or liquid food and breaks it down inside the body is called:
(A) autotrophic nutrition
(B) saprotrophic nutrition
(C) holozoic nutrition
(D) parasitic nutrition
2. The product of the fractions $\frac{2}{3}$ and $\frac{3}{5}$ is:
(A) $\frac{2}{5}$
(B) $\frac{5}{8}$
(C) $\frac{1}{5}$
(D) $\frac{6}{8}$
3. A change in which no new substance is formed and which is usually easy to reverse is called a:
(A) nuclear change
(B) chemical change
(C) physical change
(D) permanent change
4. The ratio 20 to 30, written in its simplest form, is:
(A) 20 : 30
(B) 2 : 3
(C) 3 : 2
(D) 10 : 15
5. Breaking food into smaller pieces by chewing, and then into simple substances the body can use, is called:
(A) ingestion
(B) egestion
(C) absorption
(D) digestion
6. A cow chews its cud for $\frac{3}{4}$ of an hour. Written in minutes, $\frac{3}{4}$ of an hour is:
(A) 45 minutes
(B) 75 minutes
(C) 34 minutes
(D) 30 minutes

7. A change in which one or more new substances are formed, and which is usually hard to reverse, is called a:
- (A) temporary change
 - (B) change of state
 - (C) chemical change
 - (D) physical change
8. Written as a percentage, the fraction $\frac{1}{4}$ is:
- (A) 40%
 - (B) 25%
 - (C) 4%
 - (D) 14%
9. The sharp front teeth used for cutting and biting food are called:
- (A) molars
 - (B) premolars
 - (C) canines
 - (D) incisors
10. The reciprocal of the fraction $\frac{5}{8}$ is:
- (A) $-\frac{5}{8}$
 - (B) $\frac{8}{5}$
 - (C) $\frac{5}{8}$
 - (D) $\frac{13}{8}$
11. The reddish-brown flaky substance that forms on iron left in damp air is called:
- (A) ash
 - (B) rust
 - (C) chalk
 - (D) salt
12. When iron rusts, a 50 g bar gains 10 g of extra weight. This gain, as a percentage of the original weight, is:
- (A) 50%
 - (B) 10%
 - (C) 40%
 - (D) 20%
13. The broad, flat back teeth that grind and chew food into a paste are called:
- (A) molars
 - (B) milk teeth
 - (C) incisors
 - (D) canines

14. Of a meal, the small intestine absorbs $\frac{7}{10}$ of the digested food. As a decimal, $\frac{7}{10}$ is:
- (A) 0.10
 - (B) 0.7
 - (C) 7.0
 - (D) 0.07
15. For iron to rust, two things must be present together. They are:
- (A) only dry air
 - (B) air (oxygen) and water
 - (C) oil and grease
 - (D) sunlight and salt
16. A shopkeeper buys a jar of crystallised salt for ■80 and sells it for ■100. The profit is:
- (A) ■20
 - (B) ■100
 - (C) ■180
 - (D) ■80
17. The watery liquid released into the mouth that begins the digestion of starchy food is:
- (A) sweat
 - (B) urine
 - (C) saliva
 - (D) bile
18. The value of 0.6×0.3 is:
- (A) 1.8
 - (B) 18
 - (C) 0.18
 - (D) 0.9
19. Iron is often coated with a layer of zinc to stop it rusting. This protective coating process is called:
- (A) evaporation
 - (B) crystallisation
 - (C) rusting
 - (D) galvanisation
20. To find 10% of 250, you calculate:
- (A) 2.5
 - (B) 100
 - (C) 250
 - (D) 25
21. The muscular tube that pushes the swallowed food down from the mouth to the stomach is the:
- (A) rectum
 - (B) trachea
 - (C) oesophagus
 - (D) small intestine

22. Dividing the fraction $\frac{3}{4}$ by 2 gives:

- (A) $\frac{6}{4}$
- (B) $\frac{3}{2}$
- (C) $\frac{3}{8}$
- (D) $\frac{3}{6}$

23. Pure, large crystals of a substance such as copper sulphate can be obtained from its solution by the process of:

- (A) crystallisation
- (B) galvanisation
- (C) burning
- (D) rusting

24. In a galvanised coating, zinc and iron are used in the ratio 1 : 4. If 2 kg of zinc is used, the iron used is:

- (A) 8 kg
- (B) 0.5 kg
- (C) 6 kg
- (D) 2 kg

25. Finger-like folds on the inner wall of the small intestine that greatly increase the surface for absorbing food are the:

- (A) villi
- (B) taste buds
- (C) incisors
- (D) alveoli

26. Converting the improper fraction $\frac{7}{2}$ into a mixed number gives:

- (A) $2\frac{1}{3}$
- (B) $3\frac{1}{3}$
- (C) $3\frac{1}{2}$
- (D) $7\frac{1}{2}$

27. The souring of milk to form curd is an example of a:

- (A) reversible change
- (B) change of state
- (C) physical change
- (D) chemical change

28. The fraction $\frac{3}{5}$ written as a percentage is:

- (A) 30%
- (B) 53%
- (C) 35%
- (D) 60%

29. Animals such as cows and buffaloes that quickly swallow grass and later bring it back to chew slowly are called:
- (A) carnivores
 - (B) parasites
 - (C) ruminants
 - (D) decomposers
30. A glass holds 0.25 litre of saliva-like water. The total in 4 such glasses is:
- (A) 4.25 litres
 - (B) 0.29 litre
 - (C) 0.1 litre
 - (D) 1 litre
31. The melting of ice into water is best described as a:
- (A) physical change
 - (B) burning change
 - (C) irreversible change
 - (D) chemical change
32. A pen costs ₹40. After a price rise of 25%, its new price is:
- (A) ₹50
 - (B) ₹65
 - (C) ₹45
 - (D) ₹30
33. The partly digested food that a cow brings back from its stomach to chew again is called:
- (A) bile
 - (B) saliva
 - (C) cud
 - (D) enamel
34. The sum of the fractions $\frac{1}{4}$ and $\frac{1}{2}$ is:
- (A) $\frac{1}{2}$
 - (B) $\frac{3}{4}$
 - (C) $\frac{1}{6}$
 - (D) $\frac{2}{6}$
35. The burning of a candle is a chemical change because, as it burns, the wax turns into:
- (A) more solid wax
 - (B) only melted wax that can be reused
 - (C) pure oxygen
 - (D) new substances such as carbon dioxide and water vapour

36. A digestive juice is 30% acid. In 200 mL of this juice, the amount of acid is:
- (A) 6 mL
 - (B) 60 mL
 - (C) 200 mL
 - (D) 30 mL
37. Amoeba, a single-celled animal, captures its food by putting out finger-like extensions of its body called:
- (A) villi
 - (B) pseudopodia
 - (C) cilia
 - (D) teeth
38. A ruminant's stomach has 4 chambers. If food spends $\frac{2}{5}$ of its time in the first chamber, the fraction of time spent in the OTHER chambers is:
- (A) $\frac{7}{5}$
 - (B) $\frac{2}{5}$
 - (C) $\frac{1}{5}$
 - (D) $\frac{3}{5}$
39. Photosynthesis, in which a plant makes food, is considered a chemical change because it:
- (A) only changes the leaf's colour
 - (B) forms a new substance (food)
 - (C) simply melts the leaf
 - (D) can be easily reversed
40. The percentage 50% written as a fraction in its simplest form is:
- (A) $\frac{1}{5}$
 - (B) $\frac{5}{10}$
 - (C) $\frac{50}{1}$
 - (D) $\frac{1}{2}$
41. After Amoeba surrounds its food, the food is enclosed and digested inside a:
- (A) food vacuole
 - (B) stomach
 - (C) gall bladder
 - (D) small intestine
42. The value of $\frac{1}{2}$ of $\frac{1}{2}$ is:
- (A) 1
 - (B) $\frac{1}{4}$
 - (C) $\frac{1}{2}$
 - (D) $\frac{2}{4}$

43. Cutting a sheet of paper into many small pieces is a physical change because:

- (A) no new substance is formed, it is still paper
- (B) a new substance called confetti is made
- (C) it can never be undone
- (D) the paper turns into ash

44. A cow's feed is mixed with grain and hay in the ratio 2 : 3. The total number of equal parts in the mix is:

- (A) 5 parts
- (B) 1 part
- (C) 6 parts
- (D) 23 parts

45. The largest gland in the human body, which makes bile to help digest fats, is the:

- (A) kidney
- (B) salivary gland
- (C) pancreas
- (D) liver

46. Written as a fraction in its simplest form, the decimal 0.75 is:

- (A) $\frac{3}{4}$
- (B) $\frac{75}{10}$
- (C) $\frac{7}{5}$
- (D) $\frac{1}{4}$

47. Mixing baking soda with vinegar produces lots of bubbles of gas. This fizzing tells us a ____ has occurred:

- (A) physical change
- (B) no change at all
- (C) chemical change
- (D) change of state

48. A toy marked ₹200 is sold at a 15% discount. The discount amount is:

- (A) ₹15
- (B) ₹170
- (C) ₹185
- (D) ₹30

49. The taking in of food into the body through the mouth is the step of digestion called:

- (A) absorption
- (B) egestion
- (C) ingestion
- (D) assimilation

50. A grasshopper eats $2\frac{1}{2}$ grams of leaf each day. In 4 days it eats:

- (A) 8 grams
- (B) 5 grams
- (C) $6\frac{1}{2}$ grams
- (D) 10 grams

51. When magnesium ribbon is burned, it gives a bright white light and leaves a white powder. This white powder is a:

- (A) just melted magnesium
- (B) plain water
- (C) new substance (magnesium oxide)
- (D) the same magnesium ribbon

52. Out of 25 nails, 5 have rusted. The percentage of nails that have rusted is:

- (A) 5%
- (B) 50%
- (C) 25%
- (D) 20%

53. The removal of undigested, solid waste from the body is the digestive step called:

- (A) egestion
- (B) ingestion
- (C) digestion
- (D) absorption

54. The product 0.2×0.2 is:

- (A) 0.4
- (B) 0.004
- (C) 4.0
- (D) 0.04

55. Boiling water turning into steam is a physical change because the steam can be cooled back into water. Such a change is said to be:

- (A) irreversible
- (B) chemical
- (C) reversible
- (D) permanent

56. A buffalo's milk is 8% cream. The mass of cream in 50 kg of this milk is:

- (A) 8 kg
- (B) 4 kg
- (C) 0.8 kg
- (D) 40 kg

57. In the human digestive system, most of the absorption of digested food into the blood takes place in the:

- (A) mouth
- (B) stomach
- (C) small intestine
- (D) large intestine

58. A leaf is divided so that $\frac{3}{8}$ of it is eaten by a caterpillar. The fraction of the leaf left UNEATEN is:

- (A) $\frac{11}{8}$
- (B) $\frac{5}{8}$
- (C) $\frac{3}{8}$
- (D) $\frac{1}{8}$

59. To stop iron from rusting, a common everyday method is to:

- (A) paint or grease the iron surface
- (B) sprinkle salty water on it
- (C) bury it in damp soil
- (D) leave it out in the rain

60. The simple interest on ₦500 at 10% per year for 1 year is:

- (A) ₦500
- (B) ₦100
- (C) ₦10
- (D) ₦50

61. The main job of the large intestine is to absorb from the undigested food, before the waste is removed, the:

- (A) water
- (B) bile
- (C) sunlight
- (D) oxygen

62. The fraction $\frac{1}{5}$ written as a decimal is:

- (A) 1.5
- (B) 0.15
- (C) 0.2
- (D) 0.5

63. The layer of gas high in the atmosphere that protects us from the Sun's harmful ultraviolet rays is the:

- (A) water vapour layer
- (B) ozone layer
- (C) smoke layer
- (D) rust layer

64. A salt solution is made by dissolving 25 g of salt in water to make 100 g of solution. The percentage of salt is:
- (A) 25%
 - (B) 100%
 - (C) 2.5%
 - (D) 75%
65. The pointed teeth, one on each side, used for tearing and piercing food are the:
- (A) canines
 - (B) premolars
 - (C) incisors
 - (D) molars
66. A glucose tablet weighs 0.5 gram. The number of such tablets in 3 grams is:
- (A) 3
 - (B) 6
 - (C) 1.5
 - (D) 12
67. Dissolving common salt in water is a physical change because the salt can be got back by:
- (A) evaporating the water
 - (B) burning the solution
 - (C) freezing it into ice
 - (D) adding more salt
68. If 3 oranges cost ₹15, then at the same rate 5 oranges cost:
- (A) ₹75
 - (B) ₹20
 - (C) ₹25
 - (D) ₹45
69. The hard, white, shiny substance that covers and protects the outer surface of a tooth is:
- (A) saliva
 - (B) bile
 - (C) enamel
 - (D) mucus
70. The value of $\frac{2}{3} \div \frac{1}{3}$ is:
- (A) $\frac{1}{2}$
 - (B) 2
 - (C) $\frac{2}{9}$
 - (D) 3

71. The rusting of iron is a chemical change. The new substance formed when iron rusts is called:
- (A) carbon dioxide
 - (B) zinc
 - (C) pure iron
 - (D) iron oxide
72. A shopkeeper buys a bag for ₹120 and sells it at a loss of ₹20. The selling price is:
- (A) ₹140
 - (B) ₹120
 - (C) ₹100
 - (D) ₹20
73. The first set of teeth that appear in a young child and later fall out are called:
- (A) canines
 - (B) milk teeth
 - (C) permanent teeth
 - (D) wisdom teeth
74. Rounded to one decimal place (one place after the point), the number 3.47 is:
- (A) 4.0
 - (B) 3.0
 - (C) 3.4
 - (D) 3.5
75. A change in which a new substance is formed is often shown by signs such as a change in colour, a fizz of gas, or:
- (A) becoming colder by melting
 - (B) giving out heat or light
 - (C) simply changing shape
 - (D) staying exactly the same
76. During cooking, 40% of the water in a pot of 500 mL boils away. The volume of water that boils away is:
- (A) 200 mL
 - (B) 40 mL
 - (C) 500 mL
 - (D) 300 mL
77. The flap of tissue at the back of the mouth that helps in tasting food and in mixing it with saliva is the:
- (A) liver
 - (B) windpipe
 - (C) villus
 - (D) tongue

78. A nail loses $\frac{1}{8}$ of its weight to rust each month. After it has lost rust for 3 months, the fraction of weight lost is:

- (A) $\frac{1}{8}$
- (B) $\frac{3}{8}$
- (C) $\frac{3}{24}$
- (D) $\frac{1}{24}$

79. When wood is burned, the change is irreversible. This means the ash:

- (A) can easily become wood again
- (B) cannot be turned back into wood
- (C) will cool back into a log
- (D) is just melted wood

80. The ratio of the number of incisors (8) to the number of molars (12) in an adult's set of teeth, in simplest form, is:

- (A) 4 : 6
- (B) 2 : 3
- (C) 8 : 12
- (D) 3 : 2

81. The tiny structures on the surface of the tongue that let us sense the flavour of food are called:

- (A) villi
- (B) spiracles
- (C) taste buds
- (D) incisors

82. The value of $0.9 - 0.45$ is:

- (A) 1.35
- (B) 0.5
- (C) 0.54
- (D) 0.45

83. Stretching a rubber band so it grows longer, then letting it spring back, is a:

- (A) burning change
- (B) chemical change
- (C) irreversible change
- (D) physical change

84. An item costing ₹250 is sold at a profit of 20%. The profit in rupees is:

- (A) ₹50
- (B) ₹20
- (C) ₹200
- (D) ₹300

85. A cow can live on grass because tiny helpers in part of its stomach can digest the cellulose in grass. These helpers are:

- (A) bacteria
- (B) its saliva
- (C) viruses
- (D) its own teeth

86. A fraction is multiplied by 1. The result is:

- (A) the reciprocal of the fraction
- (B) always 1
- (C) the same fraction, unchanged
- (D) zero

87. Folding a paper boat and then unfolding the paper again shows that the change is:

- (A) a burning change
- (B) chemical only
- (C) irreversible and chemical
- (D) reversible and physical

88. Half of a class are boys. Written as a percentage, the boys make up:

- (A) 25%
- (B) 100%
- (C) 50%
- (D) 12%

89. The simple sugar into which much of our starchy food is finally broken down for the body to use is:

- (A) starch
- (B) glucose
- (C) bile
- (D) enamel

90. An Amoeba splits a 0.8 mm food particle into 2 equal pieces. Each piece measures:

- (A) 1.6 mm
- (B) 0.6 mm
- (C) 0.4 mm
- (D) 0.16 mm

91. The setting of cement and the cooking of food are both examples of changes that are:

- (A) only changes of state
- (B) chemical and irreversible
- (C) physical and reversible
- (D) changes that give back the starting material

- 92.** A 40 kg sack of grain loses 10% of its mass to spoilage. The mass lost is:
- (A) 36 kg
 - (B) 400 kg
 - (C) 10 kg
 - (D) 4 kg
- 93.** The acidic juice released by the stomach, which helps digest food and kills many germs in it, contains:
- (A) bile
 - (B) lime water
 - (C) hydrochloric acid
 - (D) saliva
- 94.** Which of these fractions is the greatest?
- (A) $\frac{3}{4}$
 - (B) $\frac{1}{4}$
 - (C) $\frac{2}{3}$
 - (D) $\frac{1}{2}$
- 95.** Galvanised iron resists rust better than ordinary iron because the zinc coating:
- (A) keeps air and water away from the iron
 - (B) adds water to the iron
 - (C) turns the iron into gold
 - (D) makes the iron heavier
- 96.** The decimal 0.45 written as a percentage is:
- (A) 450%
 - (B) 45%
 - (C) 4.5%
 - (D) 0.45%
- 97.** The final step in which the absorbed, digested food is used by the body's cells to grow and release energy is called:
- (A) digestion
 - (B) assimilation
 - (C) ingestion
 - (D) egestion
- 98.** Subtracting $\frac{1}{3}$ from $\frac{5}{6}$ leaves the value:
- (A) $\frac{1}{2}$
 - (B) $\frac{4}{6}$
 - (C) $\frac{1}{3}$
 - (D) $\frac{4}{3}$

99. Dissolving sugar in a cup of tea is a physical change because the sugar:

- (A) can never be got back again
- (B) turns into a brand-new substance
- (C) is still sugar and can be recovered by evaporating
- (D) burns away into ash

100. A 200 g portion of food contains 50 g of protein. The percentage of protein in the food is:

- (A) 75%
- (B) 50%
- (C) 4%
- (D) 25%